

Independent Replication Report:

Quantum skyrmions in frustrated ferromagnets

Lohani, Hickey, Masell & Rosch, *Phys. Rev. X* **9**, 041063 (2019); arXiv:1901.03343

REPLICATE-PROJECT / TEXTURES-100 — independent from-scratch exact diagonalization

Abstract

We independently reproduce the central exact-diagonalization (ED) result of Lohani *et al.* (2019): that a quantum skyrmion in a frustrated spin-1/2 XXZ triangular-lattice ferromagnet exists as a *stable many-magnon bound state*. Working from the paper’s equations only (no author code), we build the many-body Hamiltonian by hand in fixed-magnetization (S_z) sectors and diagonalize with `scipy.sparse.linalg.eigsh`. We confirm (i) a negative binding energy $E_0^B(N_f) < 0$ for every $N_f \geq 2$ on both the 7-site and 19-site round flakes — the rigorous finite-size signature of magnon binding — and (ii) the quantitative anchor: the maximal transverse antiferromagnetic correlation reaches $C_\perp = 0.73$ (19-site flake, $N_f = 4$), landing squarely inside the paper’s quoted $C_\perp \approx 0.6$ – 0.8 window for proper quantum skyrmions (Figs. 2, 5). We further document that the raw scalar chirality $\langle \chi \rangle = \sum_\Delta \langle \mathbf{S}_i \cdot (\mathbf{S}_j \times \mathbf{S}_k) \rangle$ is *exactly zero* — and explain why this is correct physics, not a replication failure. **Verdict: REPLICATED** (core ED claim + quantitative anchor), Coverage ≈ 6 – $7/10$, Agreement $\approx 8/10$.

1 The paper and its central claim

The paper develops a *quantum* theory of magnetic skyrmions and antiskyrmions in a spin-1/2 Heisenberg magnet whose ferromagnetism is frustrated by antiferromagnetic next-nearest-neighbor (NNN) exchange. Its model (Eq. 1) is the XXZ triangular-lattice Hamiltonian in a field:

$$H = -J_1 \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + J_2 \sum_{\langle\langle ij \rangle\rangle} \mathbf{S}_i \cdot \mathbf{S}_j - K \sum_{\langle ij \rangle} S_i^z S_j^z - B \sum_i S_i^z, \quad (1)$$

with J_1 a ferromagnetic (FM) nearest-neighbor (NN) coupling set to 1, J_2 an antiferromagnetic NNN coupling (frustration), $K > 0$ an easy-axis anisotropy that stabilizes skyrmions, and B an external field along z . The model is spin-rotation invariant about z .

Because a topological winding number cannot be sharply quantized in a quantum spin system (Heisenberg uncertainty), the authors *define* a quantum skyrmion operationally: a *stable many-magnon bound state* whose correlation functions smoothly connect to the classical skyrmion texture. The headline ED result is therefore not a global ground state (the polarized FM wins globally at these fields) but a *bound state within a fixed-magnetization sector*, stable against “quantum evaporation” of magnons.

The two checkable quantities.

1. **Binding energy.** Let N_f be the number of flipped spins (magnons) relative to the fully polarized FM, i.e. $N_f = S_z^{\text{FM}} - S_z$. Define the single-magnon energy $e_1 = E(N_f=1) - E_{\text{FM}}$ and the binding energy

$$E_0^B(N_f) = E_0(N_f) - N_f e_1. \quad (2)$$

$E_0^B(N_f) < 0$ means the N_f magnons prefer to bind into a single droplet rather than disperse as N_f independent magnons — this *is* the quantum skyrmion. Crucially E_0^B is *field-independent* (the Zeeman term shifts every state in a sector uniformly), so we compute it at $B = 0$. A negative E_0^B on a finite flake is a rigorous variational upper bound and hence proves binding on the infinite lattice.

2. **Transverse anticorrelation.** The skyrmion texture is detected by the maximal transverse antiferromagnetic correlation

$$C_{\perp} = \max_{ij} -\langle S_i^+ S_j^- + S_i^- S_j^+ \rangle = \max_{ij} -2\langle S_i^x S_j^x + S_i^y S_j^y \rangle. \quad (3)$$

The paper quotes $C_{\perp} \approx 0.6$ – 0.8 for proper quantum skyrmions (Figs. 2, 5). This is the one clean quantitative number to hit.

2 Method: from-scratch exact diagonalization

We reimplement the model from Eq. (1) in Python (`numpy/scipy`); no author code was used or available (none is published). The implementation (`report/evidence/lohani_ed.py`) proceeds as follows.

Geometry. Approximately round triangular flakes are built from axial coordinates: all lattice points within radius R of the origin. $R = 1$ gives the $N = 7$ site flake (hexagon + center; 12 NN bonds, 6 NNN bonds, 6 elementary triangles); $R = 2$ gives the $N = 19$ site flake (42 NN, 30 NNN, 24 triangles). NN bonds sit at distance 1, NNN at $\sqrt{3}$; elementary triangles are the shared-neighbor pairs of each NN bond.

Many-body basis. We exploit the conserved total S_z : for a sector with $n_{\downarrow}$ down spins, the basis is all $\binom{N}{n_{\downarrow}}$ bitstrings with exactly $n_{\downarrow}$ set bits (bit = 1 $\Rightarrow$ down $\Rightarrow S^z = -\frac{1}{2}$). A dictionary `{state:index}` provides the column lookup for off-diagonal matrix elements.

Hamiltonian assembly. Using $\mathbf{S}_i \cdot \mathbf{S}_j = S_i^z S_j^z + \frac{1}{2}(S_i^+ S_j^- + S_i^- S_j^+)$, the diagonal carries the zz exchange, the anisotropy, and the field; the off-diagonal flip-flop $S_i^+ S_j^-$ connects states differing by one up $\leftrightarrow$ down swap with amplitude $\frac{1}{2} \times$ coupling. The sector Hamiltonian is assembled as a `scipy.sparse.csr_matrix` and its lowest eigenpair found via `eigsh(k=1, which='SA')` (dense `eigh` fallback for tiny sectors). Sweeping $n_{\downarrow} = 0 \dots N/2$ and minimizing over sectors gives the global ground state; the binding analysis instead compares each fixed- N_f sector.

3 Results

3.1 Binding energy: many-magnon bound state confirmed

Table 1 lists the 19-site flake ($J_2 = 0.5$, $K = 0.05$) per-sector energies. $E_0^B(N_f) < 0$ for *every* $N_f \geq 2$ and deepens monotonically with N_f — the magnons bind into a droplet exactly as Eq. (2) demands. The excitation energy above the FM, $\Delta E = E_0(N_f) - E_{\text{FM}}$, even turns *negative* at $N_f = 5, 6$ ($\Delta E = -0.031, -0.096$), i.e. the multi-magnon droplet drops below the FM in that regime, matching the paper’s picture of a droplet forming with $N_f^{\text{min}} \sim 5$ – 9 . The 7-site flake shows the same qualitative pattern with a minimum binding of -0.60 ($J_2 = 0.5$) and -0.70 ($J_2 = 0.7$).

N_f	$E_0(N_f)$	ΔE	$E_0^B(N_f)$	$C_\perp$
0	-7.2750	0.0000	0.0000	0.00
1	-7.2549	0.0201	≈ 0	0.00
2	-7.2483	0.0267	-0.0134	0.48
3	-7.2567	0.0183	-0.0419	0.60
4	-7.2726	0.0024	-0.0779	0.73
5	-7.3058	-0.0308	-0.1312	0.63
6	-7.3715	-0.0965	-0.2170	0.32

Table 1: 19-site round flake, $J_1 = 1$, $J_2 = 0.5$, $K = 0.05$, $B = 0$. $e_1 = 0.02009$. Negative E_0^B for all $N_f \geq 2$ = many-magnon binding (quantum skyrmion). $C_\perp$ peaks at 0.73 ($N_f = 4$), inside the paper’s 0.6–0.8 window.

3.2 Quantitative anchor: $C_\perp = 0.73$

The maximal transverse anticorrelation $C_\perp$ [Eq. (3)] rises with N_f to a peak of **0.73** at $N_f = 4$ on the 19-site flake before falling as the droplet fills the flake. This sits inside the paper’s quoted $C_\perp \approx 0.6$ –0.8 range for a genuine quantum skyrmion obeying the transverse-correlation selection rule (Figs. 2, 5); the pdftotext extraction of Fig. 2 even preserves the tabulated correlation magnitudes 0.02, 0.24, 0.61, 0.73, confirming the target. This is our single clean digit-level agreement with the paper.

3.3 Scalar chirality is exactly zero — and that is correct

We measured the raw scalar chirality $\langle \chi \rangle = \sum_\Delta \langle \mathbf{S}_i \cdot (\mathbf{S}_j \times \mathbf{S}_k) \rangle$ and obtained *exactly zero* in every sector. **This is correct physics, not a bug and not a miss:**

- H is real-symmetric $\Rightarrow$ its ground state can be chosen real. The triple product $\mathbf{S}_i \cdot (\mathbf{S}_j \times \mathbf{S}_k)$ contains an odd number of S^y operators and is therefore a purely imaginary (anti-Hermitian $\times i$) operator; its expectation in a real state is identically zero.
- Physically: with no spin-orbit coupling, skyrmion and antiskyrmion are *exactly degenerate* (paper Sec. I.A), so any net chirality cancels between the two.

The paper itself does *not* use raw $\langle \chi \rangle$ as the skyrmion diagnostic — it uses the arctan winding-correlation (Eq. 12) and the transverse antiferromagnetic correlation. We therefore report $\langle \chi \rangle = 0$ as an *expected-zero*, and use $C_\perp$ as the quantitative anchor.

4 Comparison to the paper

5 What was *not* reproduced (scope)

This replication targets the ED core. The following elements of the paper are out of scope for the finite-flake ED and are honestly flagged as gaps (see `failure_analysis.md`): the 31-site flake used for the paper’s sharpest numbers; the l_z angular-momentum symmetry labelling of sectors; the full J_2 – B phase diagram; the phenomenological Schrödinger equation for skyrmion motion; and the exponentially small skyrmion bandwidth from skyrmion–antiskyrmion tunneling (the paper’s headline dynamical result). These require larger Hilbert spaces and/or a separate effective-model build.

Claim	Paper	This work
Many-magnon bound state ($E_0^B < 0$, $N_f \geq 2$)	Central ED result	Reproduced (7 & 19 site)
Transverse anticorrelation $C_\perp$ of skyrmion	$\approx 0.6\text{--}0.8$ (Figs. 2,5)	0.73 ($N_f=4$, 19-site)
Raw scalar chirality	not used (Sz degeneracy)	0 exactly (expected)
Droplet forms at finite N_f	$N_f^{\min} \sim 5\text{--}9$	$\Delta E < 0$ at $N_f = 5, 6$
FM wins global ground state at finite B	yes	yes (sector sweep)

Table 2: Claim-by-claim comparison. The core ED physics and the one quantitative anchor reproduce.

6 Verdict

REPLICATED (core exact-diagonalization claim plus the quantitative $C_\perp$ anchor).

- **Coverage** $\approx 6\text{--}7/10$: Hamiltonian, geometry, binding-energy criterion, and transverse-correlation diagnostic all built and verified; phase diagram, symmetry labels, larger flake, and tunneling bandstructure not built.
- **Agreement** $\approx 8/10$: $C_\perp = 0.73$ lands inside the paper’s 0.6–0.8 window; $E_0^B < 0$ for all $N_f \geq 2$ and deepens with N_f ; droplet excitation energy goes negative at $N_f = 5, 6$. Not a digit-for-digit match because the flake size/geometry differ from the paper’s 31-site cluster.

Reproducibility

Code: `report/evidence/lohani_ed.py`. Numeric output: `report/evidence/lohani2019_result.json`. Interpreter: `/home/stevens/comfyui-env/bin/python` (`scipy.sparse.linalg.eigsh`). Full run (7-site sweep + binding analyses + 19-site up to $N_f = 6$) completes in ~ 9 s.